



School: .....Campus: .....

Academic Year: ..... Subject Name: ..... Subject Code: .....

Semester: ..... Program: ..... Branch: ..... Specialization: .....

Date: .....

## Applied and Action Learning

(Learning by Doing and Discovery)

**Name of the Experiment: Cross the Chain – Bridge or Interoperability Demo**

**\*Coding Phase: Pseudo Code / Flow Chart / Algorithm**

1. Start the experiment in Remix IDE.
2. Create two contracts:
  - LockContract → simulates the source blockchain where tokens are locked.
  - ReleaseContract → simulates the destination where tokens are released.
3. Compile both contracts using Solidity compiler version 0.8.x.
4. Connect MetaMask to Ethereum Sepolia Testnet and ensure test ETH balance.
5. In Remix, select Injected Provider – MetaMask as the environment.
6. Deploy both contracts on Sepolia network.
7. Lock tokens by sending Ether using lockTokens() from an account (User A).
8. Observe the event log “TokensLocked” emitted from the contract.
9. Call the releaseTokens() function in the second contract (ReleaseContract) using data from the first contract (sender, amount).
10. Check balances on both contracts to verify lock and release.
11. Record observations and transaction results.
12. End the simulation.

### \* Software used:

- Remix IDE – For writing and deploying smart contracts.
- MetaMask Wallet – To manage accounts and connect to the blockchain.
- Ethereum Sepolia Testnet – Used as the simulation environment.
- Solidity Compiler v0.8.x – For contract compilation.
- Injected Provider – MetaMask – For contract deployment via Remix.

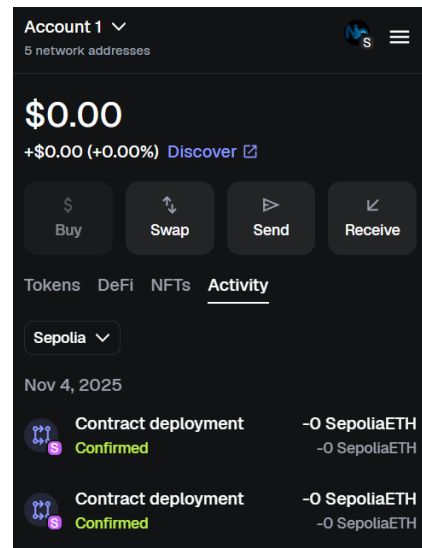
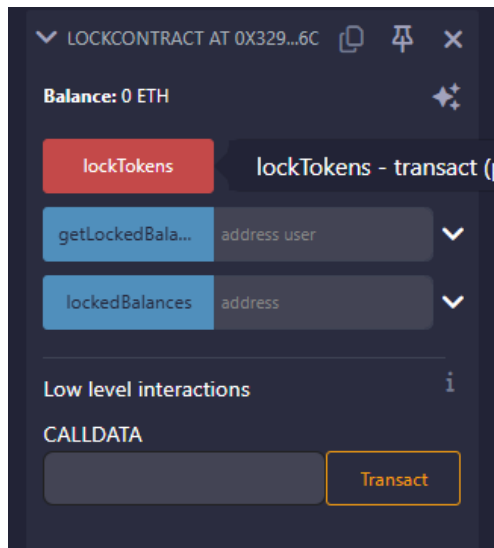
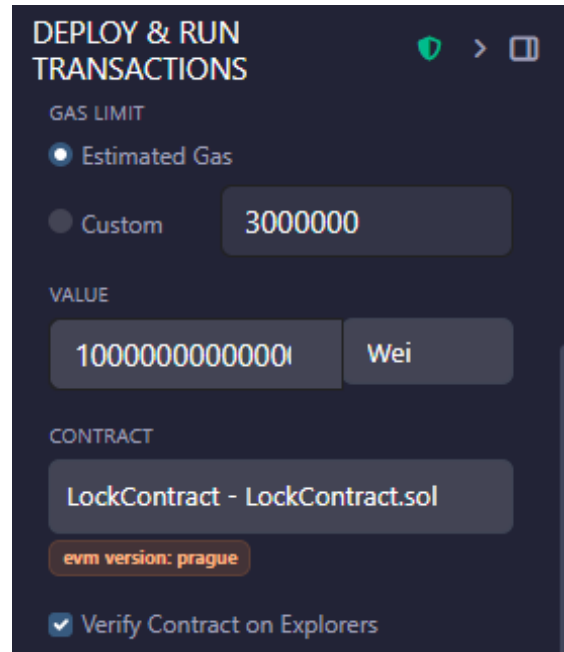
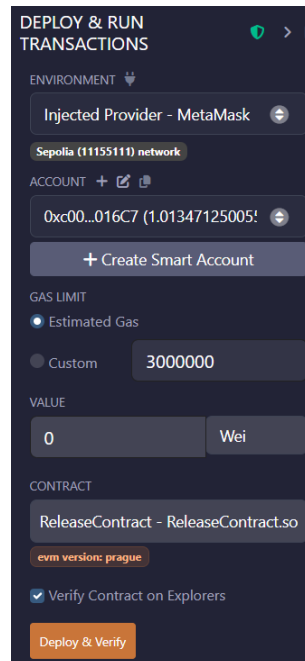
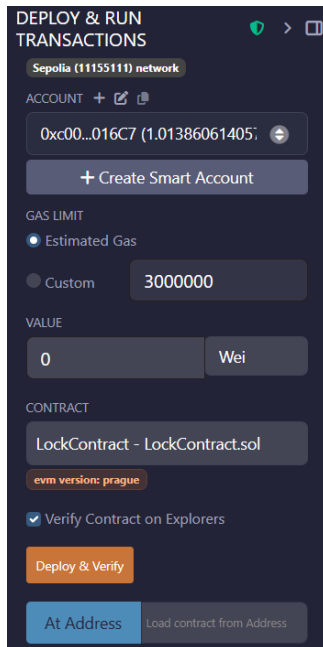
Page No.....

**\* As applicable according to the experiment.  
Two sheets per experiment (10-20) to be used.**

## \* Testing Phase: Compilation of Code (error detection)

NO ERROR

## \* Implementation Phase: Final Output (no error)



```
getLockedBalance(sender) → 0.01 ETH
getReleasedBalance(sender) → 0.01 ETH
```

Page No.....

\* As applicable according to the experiment.  
Two sheets per experiment (10-20) to be used.

### \* Observation :

The experiment demonstrates:

- Asset locking and event emission on one chain.
- Authorized release on another chain.
- Matching balances for locked and released assets.

## ASSESSMENT

| Rubrics                                                     | Full Mark | Marks Obtained | Remarks |
|-------------------------------------------------------------|-----------|----------------|---------|
| Concept                                                     | 10        |                |         |
| Planning and Execution/Practical<br>Simulation/ Programming | 10        |                |         |
| Result and Interpretation                                   | 10        |                |         |
| Record of Applied and Action Learning                       | 10        |                |         |
| Viva                                                        | 10        |                |         |
| <b>Total</b>                                                | <b>50</b> |                |         |

**Signature of the Student :**

**Name :**

**Regn. No. :**

**Signature of the Faculty :**

Page No.....

*\* As applicable according to the experiment.  
Two sheets per experiment (10-20) to be used*